

Sandra Ilievska

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PROFILE

Full-stack engineer with real production experience across backend (.NET, Laravel) and frontend (Vue, React, TypeScript). I also do AI research on the side – 2 papers accepted, 3 more under review, spanning cultural heritage AI and network automation safety. Recently one of 290 people picked from 2,600+ applicants for a top European machine learning summer school. Looking for a full-stack or backend role where I can keep building real systems and stay close to applied AI.

EXPERIENCE

ELIKOSOFT

Nov. 2025 – Apr. 2026

Skopje, North Macedonia

Software Developer

Feb. 2026 – Apr. 2026

- Built a role-based access control system for 5 user roles, with module-level permissions, using the ABP framework and .NET
- Added workflow automation, filtering, and search to core ERP modules used daily by internal teams
- Refactored legacy backend code across 3 modules, cutting complexity and making the code easier to maintain
- Sped up data access by rewriting slow SQL queries with indexed joins
- Wrote parts of the frontend with Razor, TypeScript, and JavaScript

Software Engineer Intern

Nov. 2025 – Feb. 2026

- Created and tested REST API endpoints for internal ERP workflows, using ASP.NET
- Worked on database schema design and query optimization on SQL Server
- Worked directly in the production codebase on filtering and search features

EDUCATION

FCSE, Ss. Cyril and Methodius University

Oct. 2022 – Present

Software Engineering

Brainster Academy

Skopje, North Macedonia

Feb. 2025 – May 2026

Full-Stack Development Program – completed with Distinction, 100/100

Skopje, North Macedonia

SkillUp Academy

Sep. 2021 – Mar. 2022

Frontend Development Program

Veles, North Macedonia

CONFERENCES & RESEARCH

ICT Innovations 2026 — International Conference

Under Review: Overview of Cultural Heritage Initiatives for Digitization of Balkan Traditional Embroidery

- Led a structured literature review of digitization efforts for Balkan folk embroidery, covering institutions in 5 countries (North Macedonia, Serbia, Bulgaria, Kosovo, Albania)
- Assessed the scale, access, and licensing of existing digital collections (Europeana, Wikimedia Commons, national museums)
- Found a consistent gap: large physical holdings, almost no searchable digital catalogs – worst in North Macedonia, which has no open dataset at all

TELFOR 2026 — International Conference

Under Review: From Thread to Tensor — A Digitization-to-Generation Pipeline (Part 2)

- Implementation sequel to the survey above – moved the project from literature review to a working system
- Built EtnoMK, an image-similarity search tool (DINOv2 embeddings + pgvector) that classifies archival embroidery photos by sub-region
- Scoped the next two stages (motif extraction, pattern generation) against existing GAN/diffusion research, with concrete accuracy targets to benchmark against

Tech: Python, DINOv2, pgvector, PostgreSQL

TELFOR 2026 — International Conference

Under Review: LLM Agent Orchestration for Network Change Automation

- Designed a guardrail architecture for safe LLM-agent network changes: structured JSON validated by deterministic checks, plus human-in-the-loop approval
- Set up an identical 10-step network-provisioning pipeline across three agent platforms – LangGraph complete and evaluated, n8n in progress, Langflow planned next
- Found that a local LLM (qwen2.5:7b) had a 0% autonomous success rate across 40 safe test runs, split between hallucination and model over-caution

Tech: Python, LangGraph, n8n, Langflow, Ollama (open-weight LLMs)

CIIT 2026 — International Conference on Informatics and Information Technologies

Presented Project: Automated Company Ranking and Segmentation via Open Data Registries

- Built a ranking and enrichment system covering 5,967 companies across 10 sectors from North Macedonia's public registries, scored by a custom Digital Readiness Index
- Lifted email coverage from 5.5% to ~20% with a 3-strategy enrichment pipeline, plus automated duplicate and inactive-record detection
- Added AI agents with structured output to generate personalized, sector-specific outreach offers and target lists

Tech: Laravel, Filament, Symfony BrowserKit, OpenAI API, SQLite, Pest

ICT Innovations 2025 — International Conference

Presented Project: TravelSage: A Database-Driven Platform for Personalized Travel Planning

- Created a full-stack travel recommendation system combining user preferences, weather data, and traveler feedback
- Designed relational data models for destinations, activities, and user-driven recommendations
- Wrote backend logic for data-driven decision making and ranking of destinations

Tech: Laravel, PostgreSQL, Blade, Tailwind CSS

ADDITIONAL ACTIVITIES

EEML 2026 — Eastern European Machine Learning Summer School | *Cetinje, Montenegro*

- Selected as 1 of ~290 participants out of 2,600+ applicants worldwide
- Attended sessions on geometric deep learning, embodied AI, diffusion models, continual learning, and AI for biology, taught by researchers from Google DeepMind, MIT, NYU, EPFL, and Oxford

Intelligent System for Plant Disease Detection | *Project Manager — University Project*

- Led a university team building an AI system to detect plant diseases from photos, across mobile, web, backend, and AI components
- Managed the backlog and task assignment in Jira, ran progress check-ins, and organized team meetings
- Helped unblock technical issues across a Flutter mobile app, React web frontend, Python/FastAPI backend, and a TensorFlow/PyTorch image-analysis module

Team Europe 2026 Cybersecurity Bootcamp | *Brussels, European Parliament*

- Selected from FINKI's cybersecurity students as one of two first representatives of North Macedonia, joining 70+ peers from across Europe
- Completed intensive training in web security, cryptography, digital forensics, and CTF challenges

Shared Horizons Youth Bootcamp | *Belgrade, Western Balkans*

- Selected to represent North Macedonia in a 5-day program with participants from across the Western Balkans, focused on leadership, dialogue, and social cohesion

Hackathons, CTF Competitions & Community Events

- Regular participant in backend-focused hackathons and international CTF competitions (cryptography, reverse engineering); won team challenge at Women4Cyber North Macedonia digital forensics workshop
- Frequent attendee of AI and tech industry events, including the StartupRevolution.ai Summit

PROJECTS

Project Manager Platform | *Laravel, Vue 3, Inertia.js, TypeScript, Filament*

- Created a full-stack task and project management system with role-based access control and workflow automation
- Set up the modular backend architecture, admin dashboard, and team collaboration features

AI Tools Digital Library | *Django, Python, Bootstrap, Chart.js, SQLite*

- Put together a searchable catalog of AI tools with filtering, bookmarking, and analytics dashboard; used as an admin-managed content platform
- Added a service-layer architecture with optimized search indexing, separating business logic from data access

Hear & See Safe (Accessibility App) | *Flutter, Dart, TensorFlow Lite*

- Multilingual accessibility application for children with visual and hearing impairments
- Integrated voice interaction, learning modules, and accessibility-first UI design

LLM Intent-Based Networking Simulator | *Python, NetworkX, OpenAI API*

- Simulation system translating natural-language intents into network flow rules on virtual topology
- Combined regex rules with optional LLM-based interpretation for a hybrid parsing approach
- Related to ongoing research paper on LLM agent orchestration (under review)

TECHNICAL SKILLS

Backend: .NET / ASP.NET / ABP, Laravel / PHP, Django

Frontend: TypeScript, Vue.js, React, Flutter, Razor

Databases: PostgreSQL, MySQL, SQL Server, SQLite

AI/ML: OpenAI API integrations, TensorFlow Lite (mobile)

Tools: Git, GitHub Actions, Azure DevOps, Jira